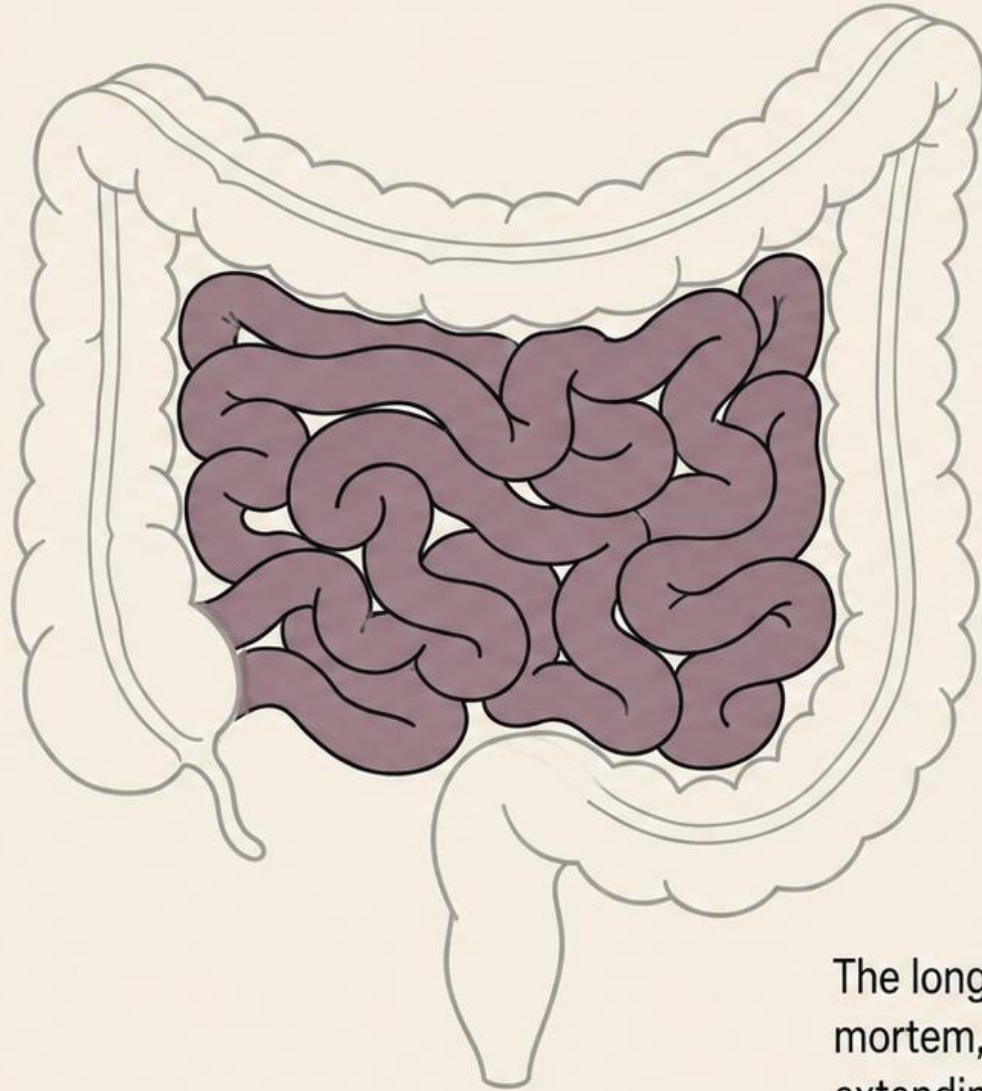


The Small Intestine: Anatomy, Ecosystem and ROP Approach

Chapter 14 — The systemic, mechanical, neurological and immune crossroads.

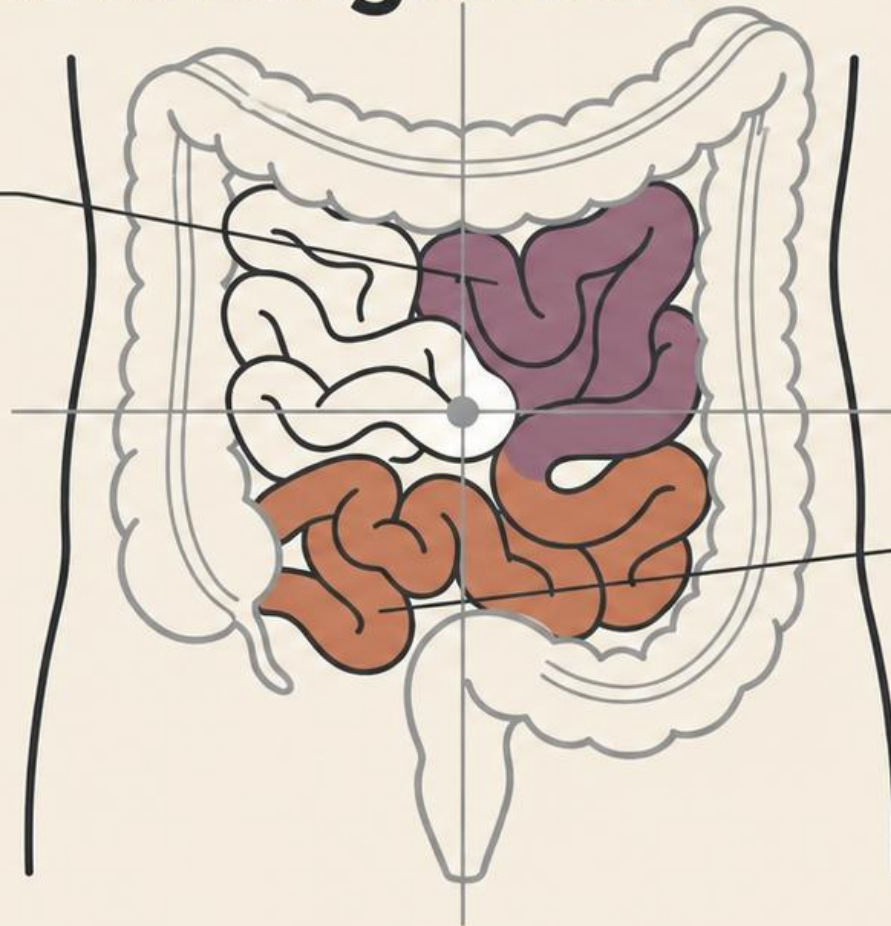


The longest organ in the body (6 to 7 meters post-mortem, shortened in vivo by muscle tone), extending from the pylorus to the ileocecal valve.

Topography and intra-abdominal arrangement

Jejunum

- Upper left quadrant
- Overlies the descending colon
- Vascular orientation and loops predominantly horizontal

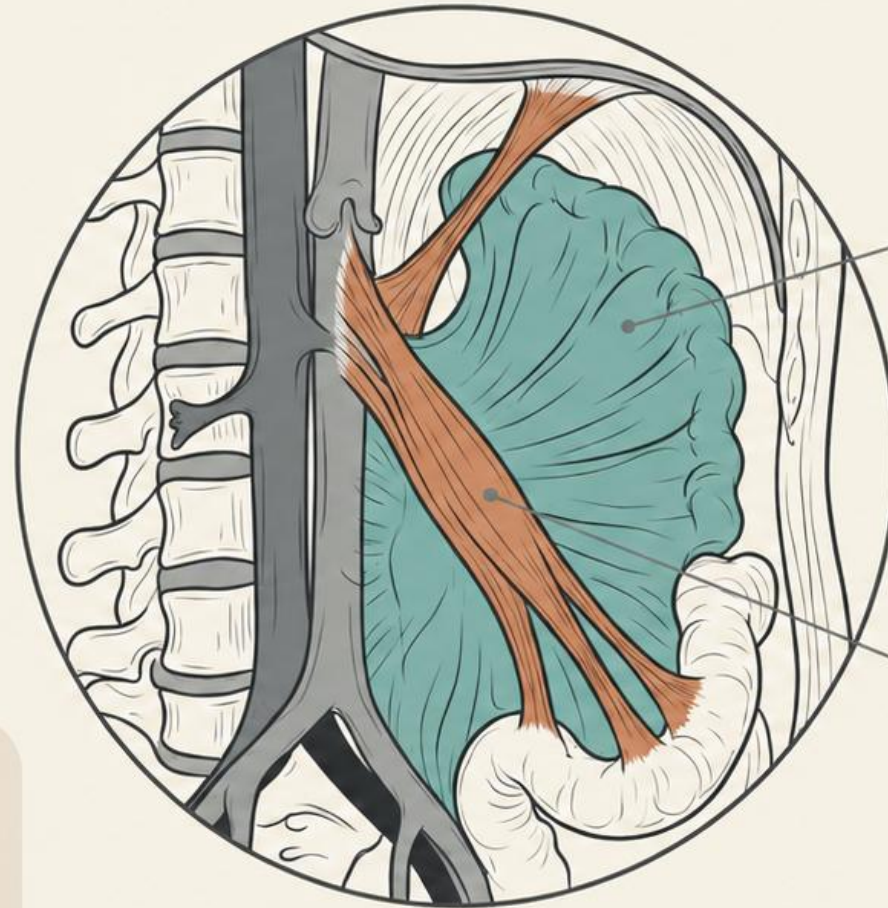


Ileum

- Lower right quadrant
- Leaves the ascending colon free
- Vascular orientation and loops predominantly vertical

Cadaveric length: 6 to 7 meters. Lumen: 2 to 3 cm in diameter.
Extremely mobile intraperitoneal organ.

Mechanical Anchoring: Mesentery and Treitz Muscle



• Mesenteric root

Attachment line 16 to 18 cm long, crossing the umbilicus at L3-L4. Ensures stability and carries vessels/nerves.

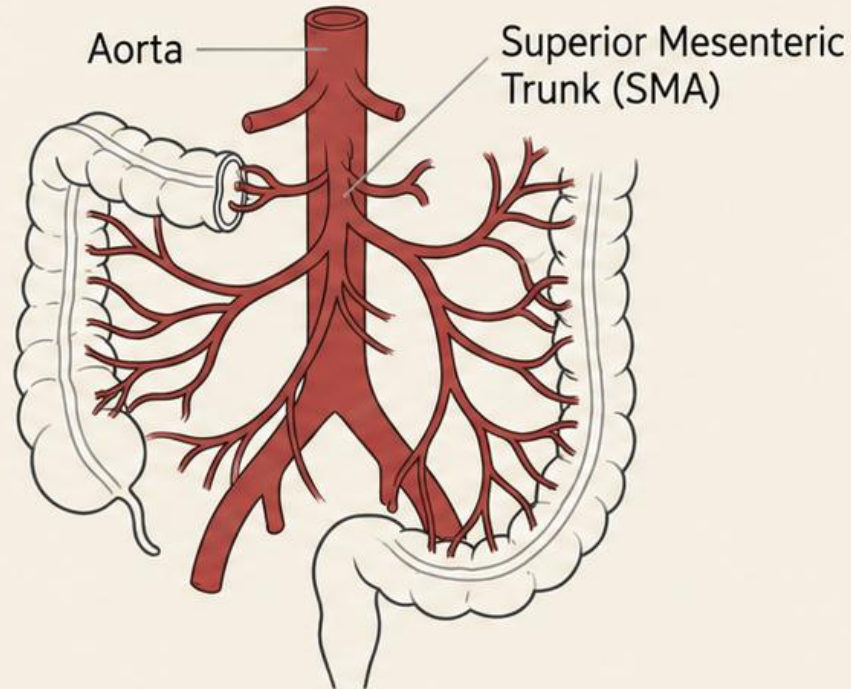
• Treitz muscle

Attaches to the right crus of the diaphragm. Maintains the duodenojejunal angle and ensures longitudinal tension.

Aorto-Mesenteric Clamp Syndrome

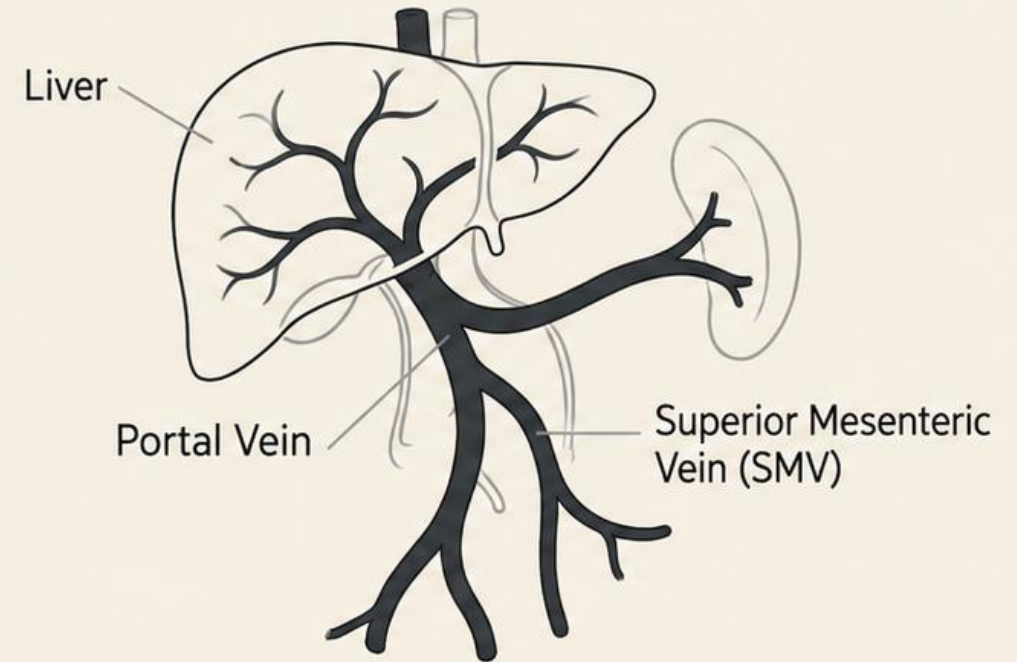
A loss of longitudinal tension can lead to compression of the 4th part of the duodenum (nutcracker syndrome), delaying emptying and promoting gastro-duodenal reflux.

The axis of irrigation and drainage: Superior Mesenteric Trunk



Arterial Supply (Superior mesenteric artery)

- Origin: T12–L1 disc (below the celiac trunk)
- Supplies the small intestine, the ascending colon, and the proximal 2/3 of the transverse colon



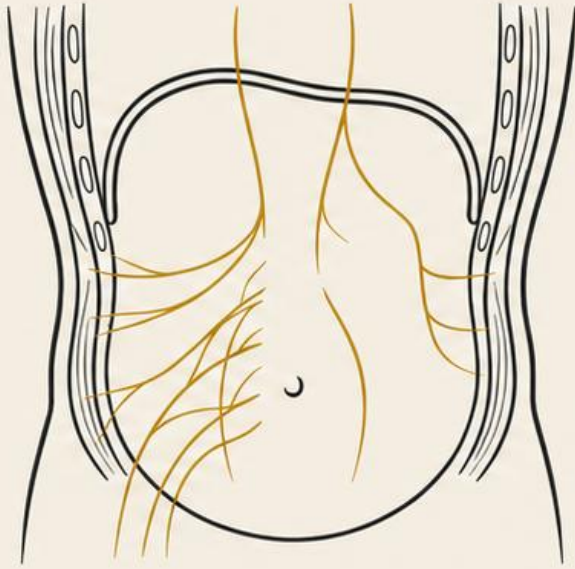
Venous Drainage (Portal System)

- Carries venous blood rich in absorbed nutrients directly to the liver for detoxification and metabolism.

Practical Note: The pulse of the SMA can be felt at the level of the 3rd part of the duodenum, to the right of the umbilicus.

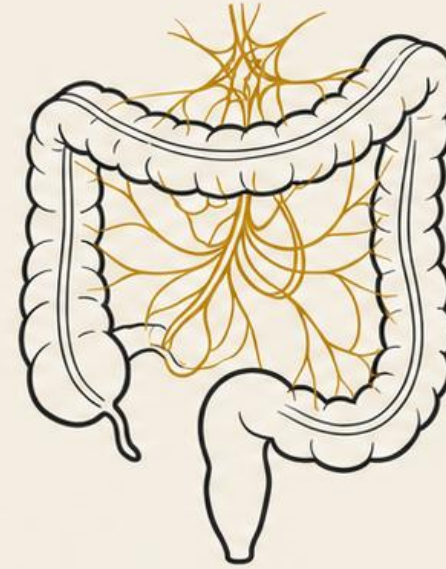
The double neurological reality of the peritoneum

Parietal Peritoneum (Somatic Innervation)



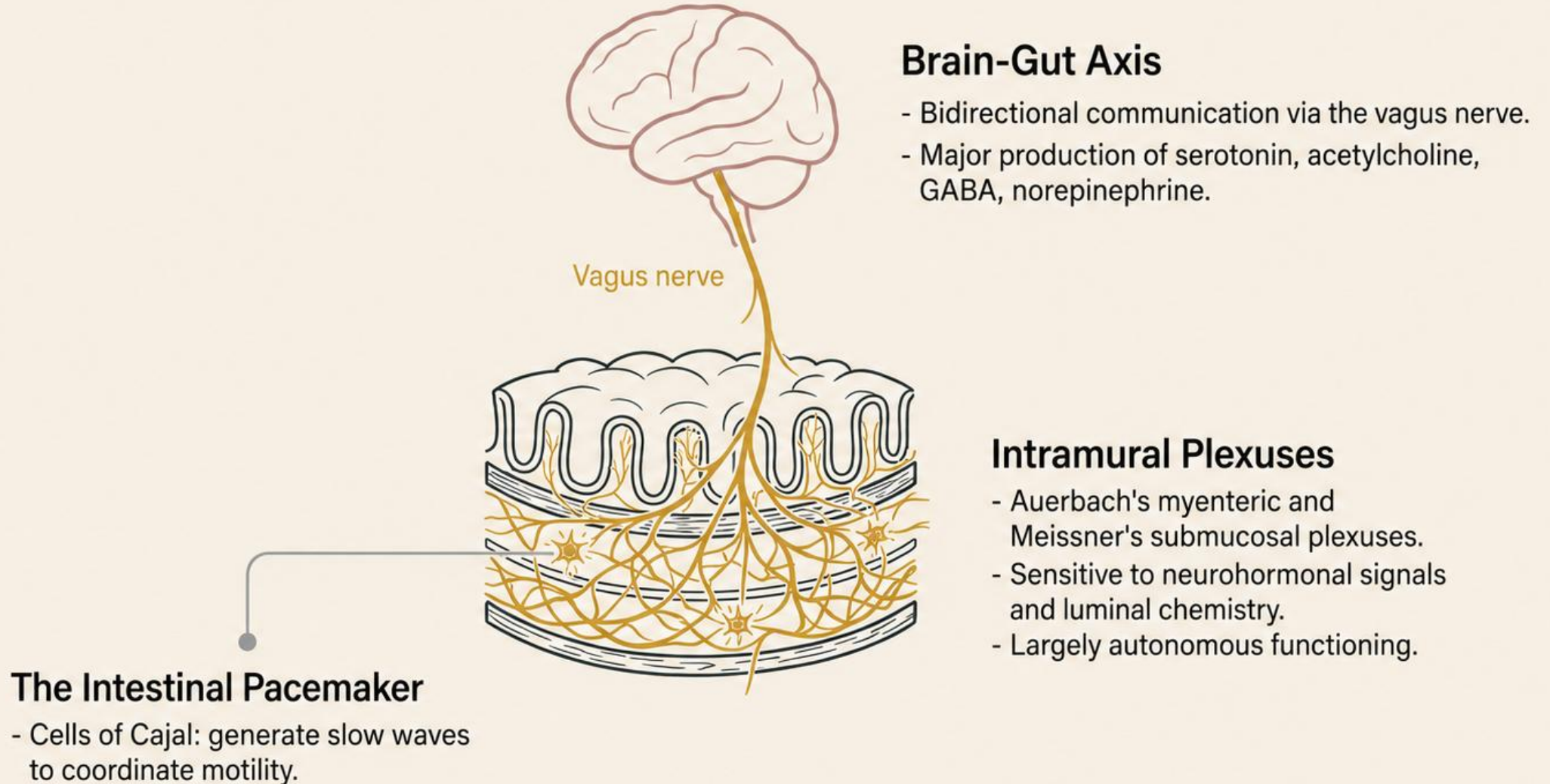
- Pathways: Phrenic nerves, last 6 intercostal nerves, lumbar plexus.
- Sensitivity: Sharp (pain, pressure, temperature).
- ROP implication: Projection of peritoneal pain toward the osteo-musculo-articular system (cervical, lumbar, scapular).

Visceral Peritoneum (Autonomic Innervation)



- Pathways: Sympathetic system (T8-T11, splanchnic nerves) and parasympathetic system (vagus nerve).
- Sensitivity: Low for touch/incision. Stimulated by stretching (distension) and chemical stimuli.
- ROP implication: The sympathetic system inhibits peristalsis (stress); the vagus promotes it.

The Second Brain: Autonomy of the Enteric Nervous System (ENS)



The "Second Brain": Enteric Nervous System

Intramural Plexuses

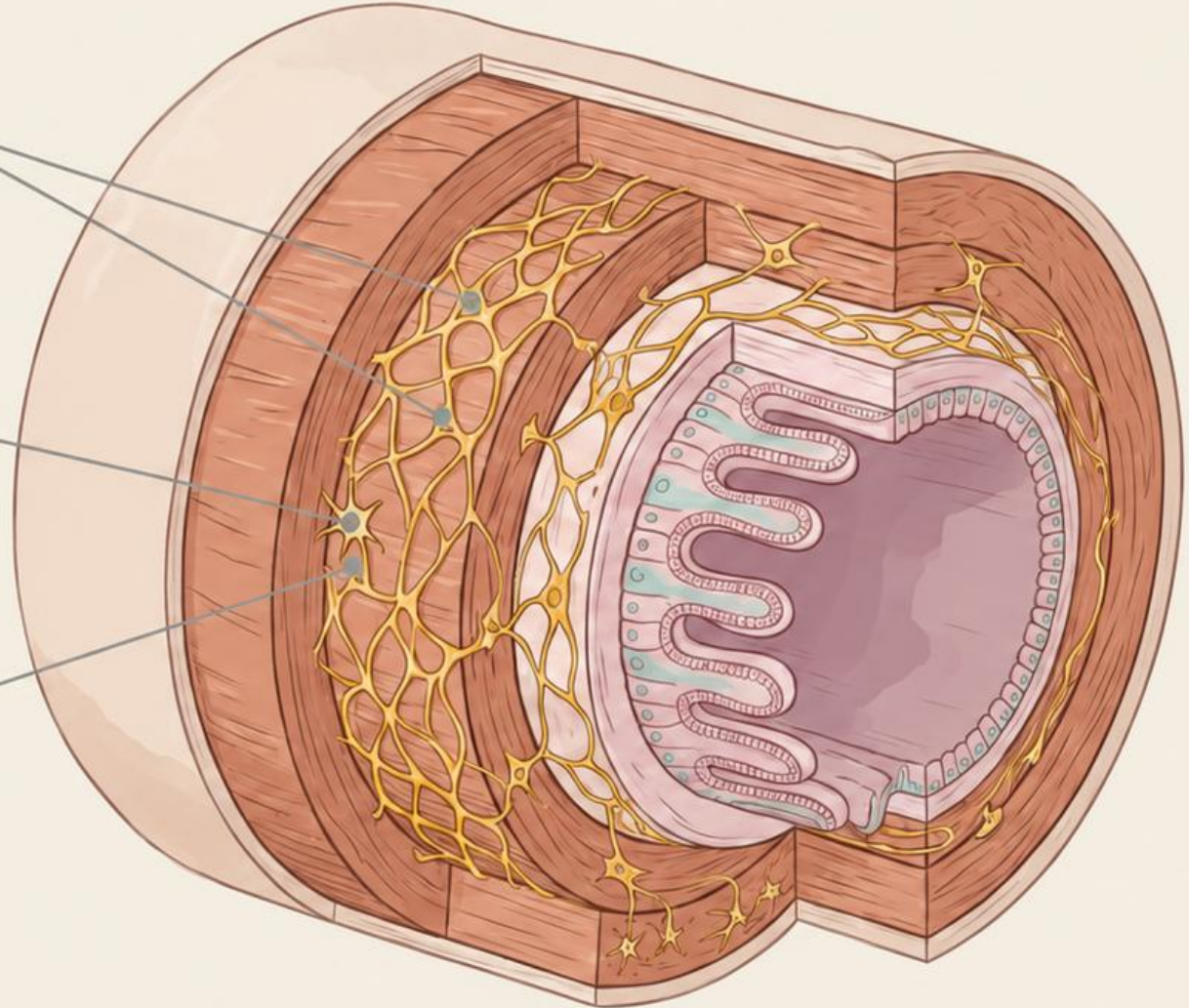
Auerbach's myenteric and Meissner's submucosal plexuses. Provide near-autonomous regulation of digestive functions.

Cajal Cells

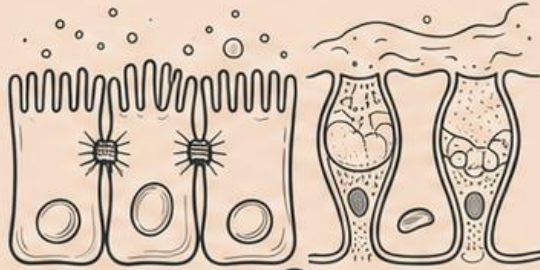
The intestinal pacemaker. Generate slow waves that coordinate motility independently of chyme-induced peristalsis.

Neurochemical Production

Major synthesis of serotonin (mood regulation shared with the brain), acetylcholine, norepinephrine, and GABA.

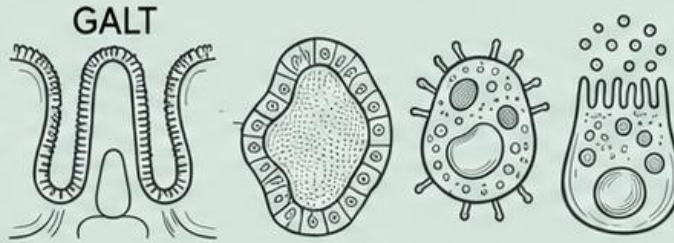


The Intestinal Ecosystem: The Tripod of Digestive Health



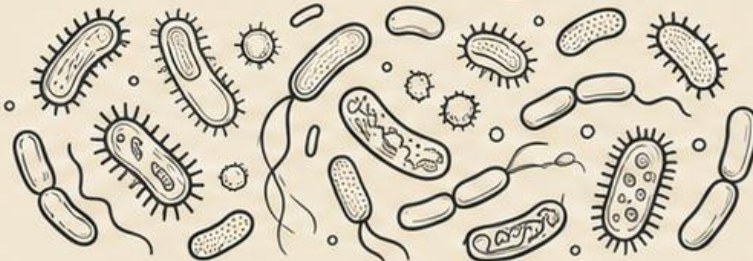
1. Intestinal Mucosa (Interface)

- Enterocytes with desmosomes (tight junctions).
- Goblet cells (production of protective mucus).



2. Immune System (Defense)

- GALT (Gut Associated Lymphoid Tissue) and Peyer's patches.
- Mast cells and Paneth cells (antimicrobial peptides).



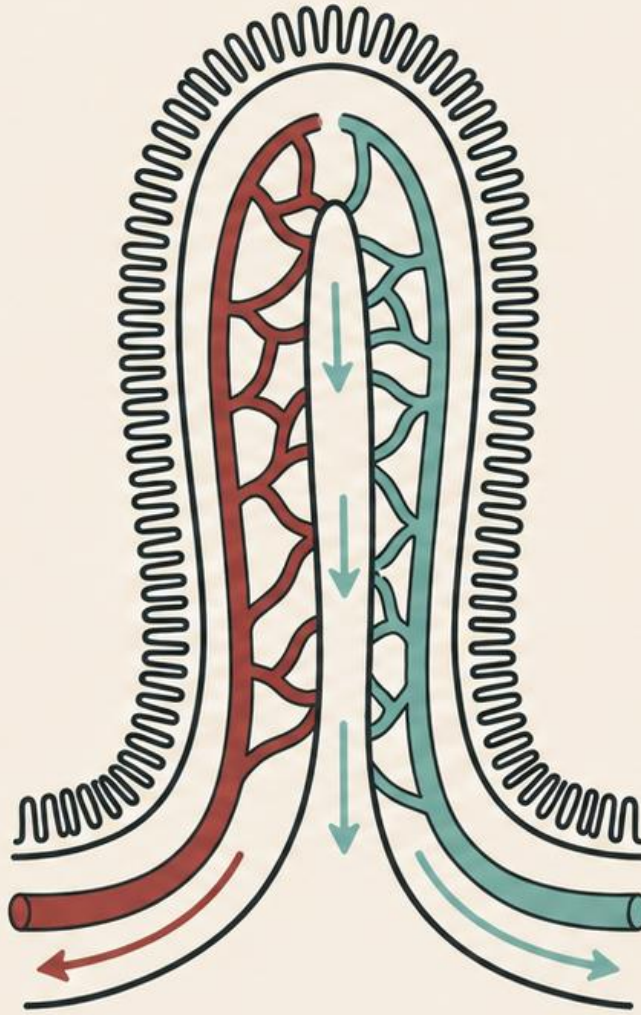
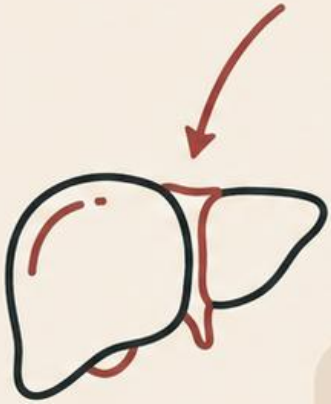
3. Microbiota (Symbiosis)

- ~38 trillion bacteria per individual.
- Unique signature from birth. Contributes to immunity, digestion, and the gut-brain axis.

The Dual Pathway of Micronutrient Absorption

Blood Pathway (Capillaries)

- Absorbed: Water, simple sugars, amino acids, water-soluble vitamins.
- Destination: Transported directly to the liver via the portal system.



Lymphatic Pathway (Lacteals)

- Absorbed: Fats, long-chain proteins, fat-soluble vitamins (A, D, E, K).
- Destination: Lymphatic network -> Cisterna chyli -> Thoracic duct.



Relevance in ROP: A foot reflex action on the liver and the lymphatic system qualitatively supports absorption and immunity.

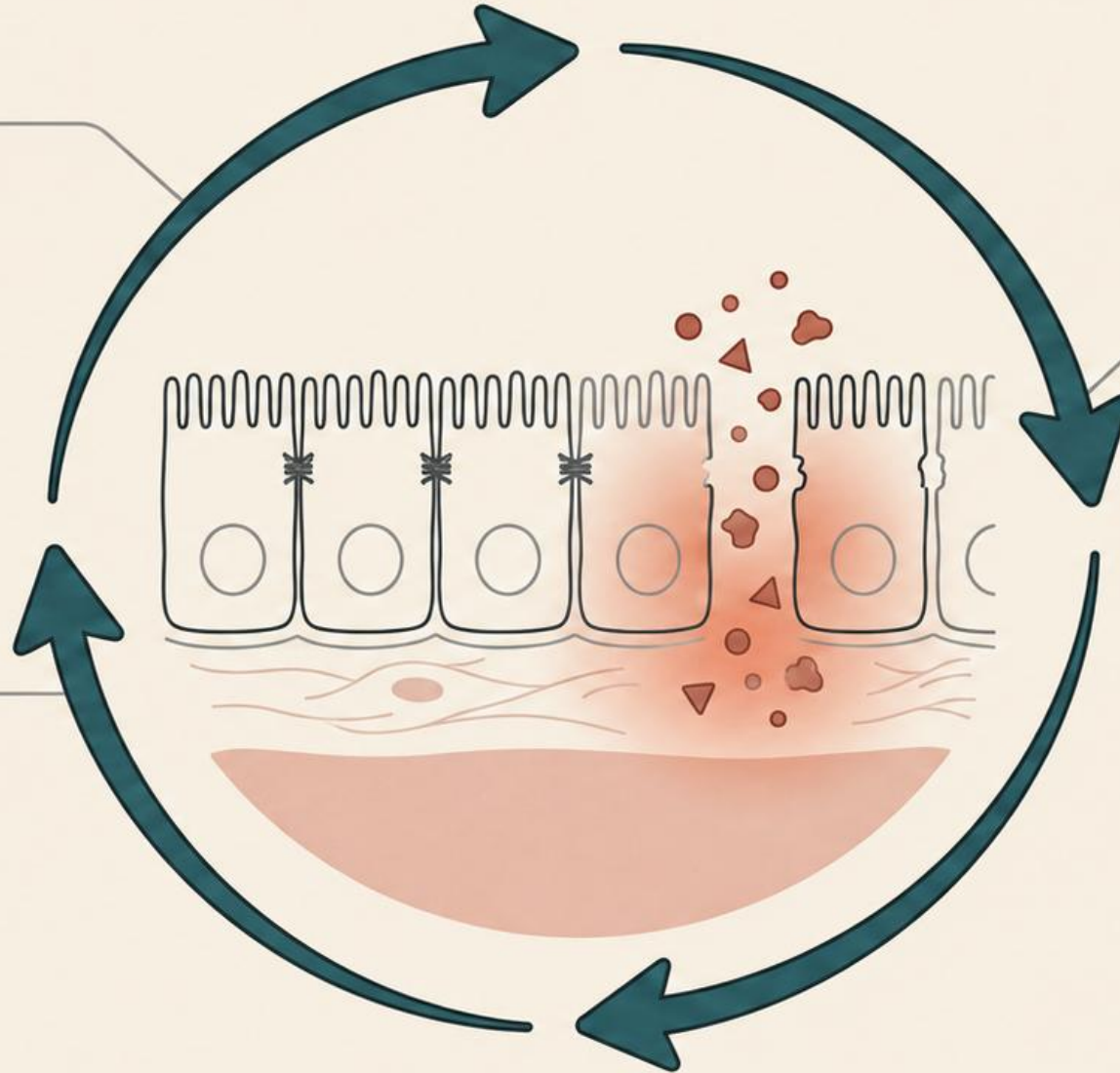
The Spiral: Dysbiosis and Hyperpermeability

Triggers

Chronic stress
(sympathetic
vasoconstriction),
ultra-processed foods,
NSAIDs, antibiotics.

Hyperpermeability

Alteration of tight junctions.
Passage of toxins and
incompletely degraded
proteins, triggering
inflammation.



Systemic Consequences

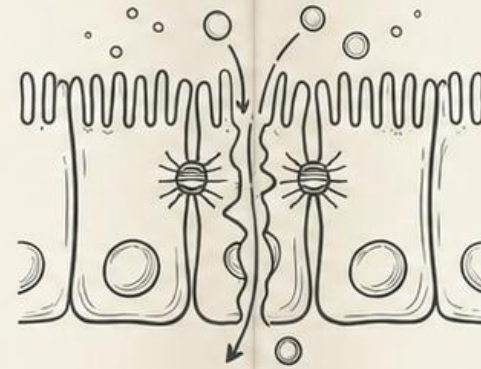
Hepatic overload
(Seignalet's clogging
theory). The saturated
liver redirects waste
to the tissues, causing
osteo-articular pain,
tendon pain, and
allergies.

Pathology I: Intestinal Hyperpermeability (Leaky Gut)

Causes (Triggers)

- Chronic stress
(vasoconstriction caused by the sympathetic nervous system)
- NSAIDs / Antibiotics
- Ultra-processed diet
- Extreme endurance sports

Mechanism (Breakdown)



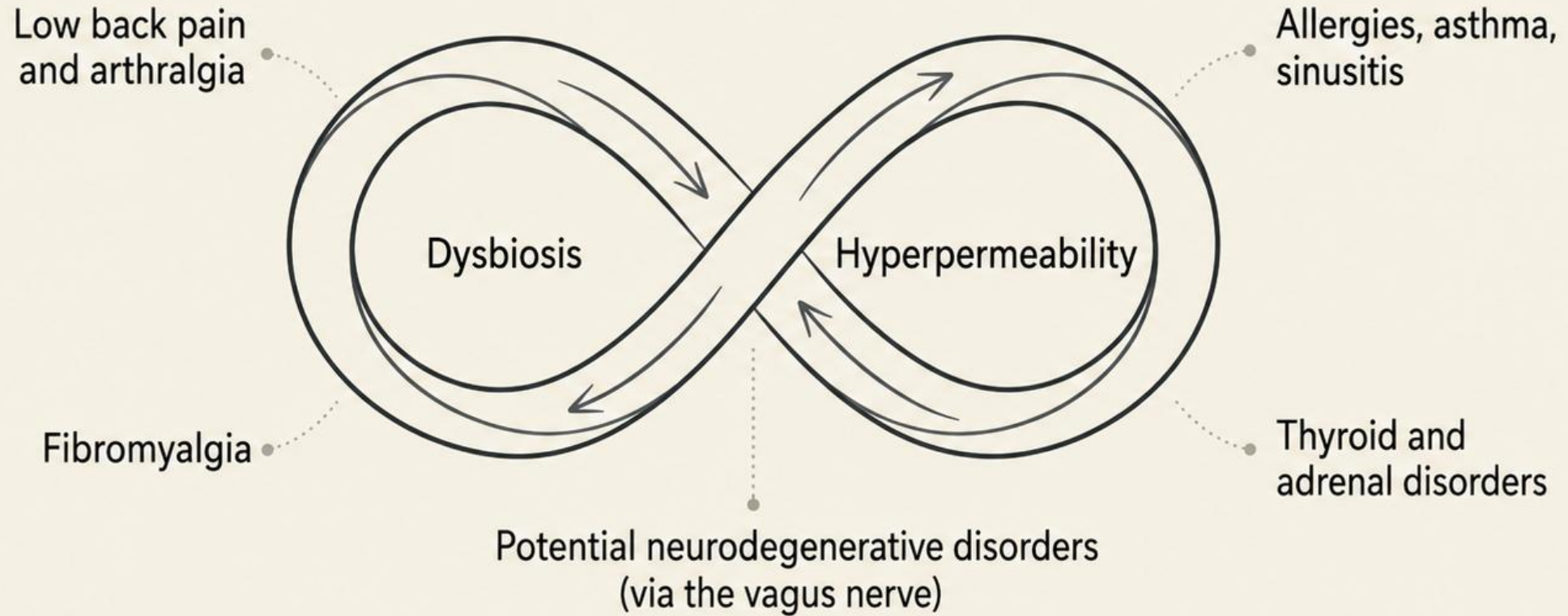
Alteration of intercellular junctions → Passage of insufficiently degraded macromolecules.

Consequences (Systemic)

- Immune activation
(allergies, autoimmune diseases).
- Hepatic overload
(Dr. Seignalet's 'clogging' theory: deposition of toxins in the muscles, joints, and brain).

ROP relevance: In the presence of osteo-articular pain without obvious trauma, always suspect a link with visceral dysfunction.

Pathologiy II: Dysbiosis and its manifestations



Warning signs: Bloating, aerocolia, alternating diarrhea/constipation, heartburn, abnormal craving for sugar/raw meat, bad breath.

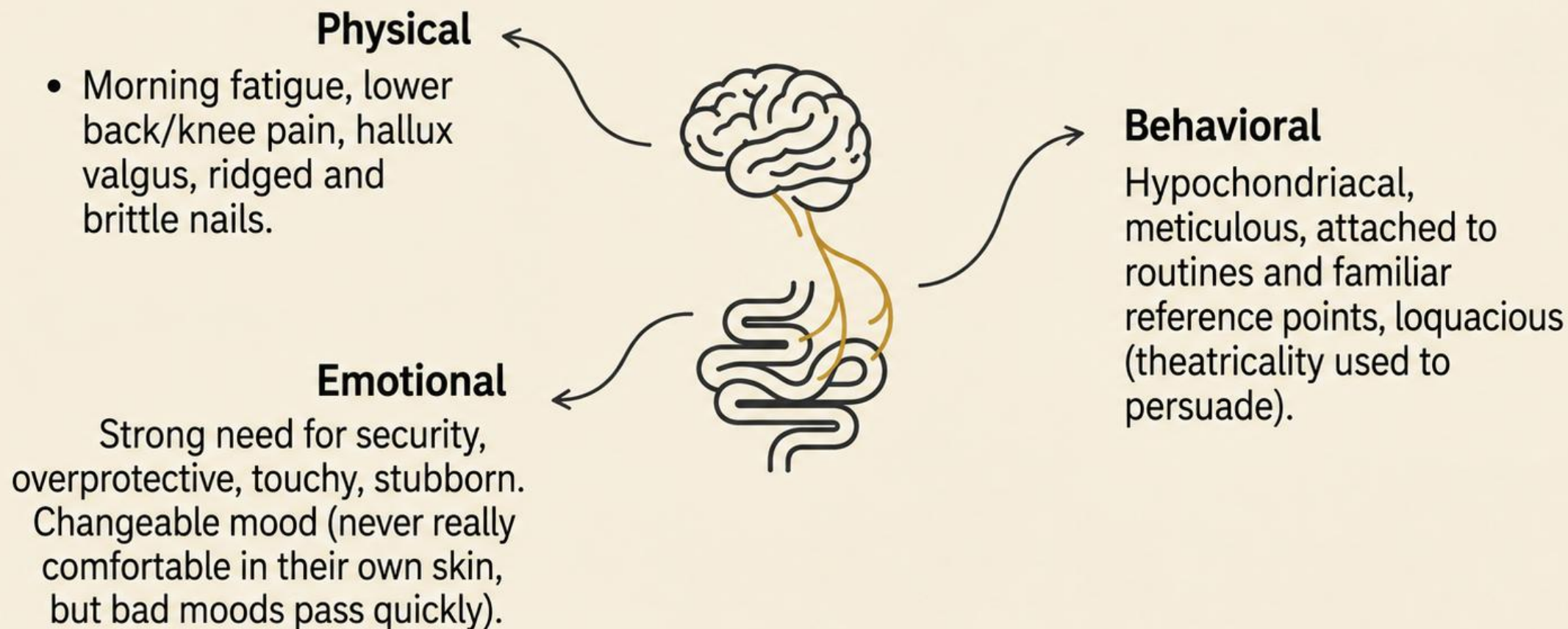
Note: Correlation $\neq$ Causality (Prof. Sokol): the two disorders coexist and mutually reinforce each other.

Warning Signs and Diagnosis of Exclusion

Immediate Medical Referral Is Imperative in the Presence of:

- - Fever, red or black blood in the stools.
- - Significant and unexplained weight loss.
- - Troisier's node (left jugulo-subclavian junction): oncologic suspicion.
- - Ice-cube sign: suggestive of ascites.
- - Irreducible strangulated inguinal hernia.
- - Obstruction / Ileus: Cessation of stool and gas passage. May be mechanical (tumor, adhesion) or paralytic (atony due to peritoneal irritation).

Viscero-Emotional Profile: The Intestine Person



Note: Highly hormone-dependent (especially in women). The small intestine is the primary receiver-transmitter of emotional tensions, promoting sympathetic hyperactivity.

ROP Strategy 1: Regulation of the Nervous System and Somatic Links



Viscero-Somatic Relationships

- Target vertebral fixations: Th10 to Th12 (and associated ribs), L1 and L2.



Foot Reflex Zones - Adaptation Syndrome

- Vagus Nerve: Cranial, cervical and abdominal territory (esophageal hiatus, cardia).
- Sympathetic System: Lateral vertebral ganglion chain (Th8-Th10) and diaphragmatic crura.
- Solar (Celiac) Plexus & Lumbar Plexus.
- Limbic System Induction: Limbic Brain-Intestine balancing technique to regulate the viscerο-emotional axis.

ROP Support for Functional Disorders



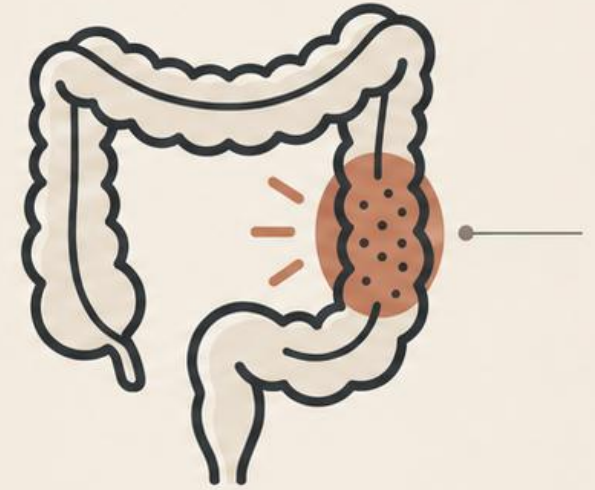
Enteroptosis

The downward pull of the intestinal mass leads to venous/lymphatic stasis, nervous spasms, and pressure on the bladder (urinary leakage). ROP aims to restore tension in the mesenteric root.



Paralytic Ileus

Motility inhibition due to sympathetic dominance (post-surgery, shock, hypokalemia). No distension pain. The induction aims to reactivate the vagus nerve.



Crohn's Disease

Genetic, dysbiotic, and emotional components. The ROP approach helps relieve abdominopelvic pain and supports better bowel transit.

ROP Intervention Summary: Small Intestine Protocol

Step 1: Listening and Induction (Limbic System)

Action: One thumb on the Small Intestine area, the other on the Limbic Brain area.

Objective: Soothe the viscerο-emotional load (stress, anxiety, need for safety) and reactivate vagal tone.



Step 2: Structural Release (Viscero-Somatic Relationships)

Action: Treatment of vertebral and costal fixations.

Target zones: T10-T12 junction (sympathetic) and L1-L2 (lumbar plexus, root of the mesentery).



Step 3: Locoregional Treatment (Visceral)

Action: Release of suspension tensions and stimulation of peristalsis.

Target zones: Diaphragm (Treitz), Stomach (reflex), Root of the mesentery (longitudinal tension), and Jejunum/Ileum mapping.

The ROP approach restores mobility, innervation, and the ecosystem: a global response to the maxim "All disease begins in the intestine".